A Quasiparticle Excitation Ansatz for 2D Isometric Tensor Network States

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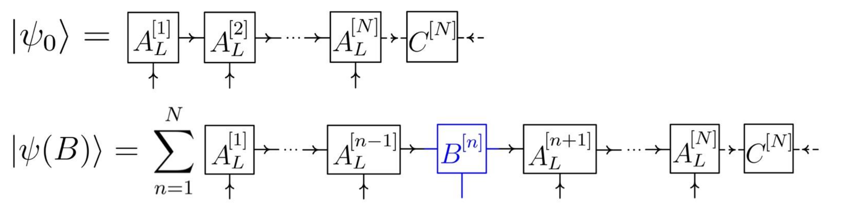
MPS

Quasiparticle excitation ansatz

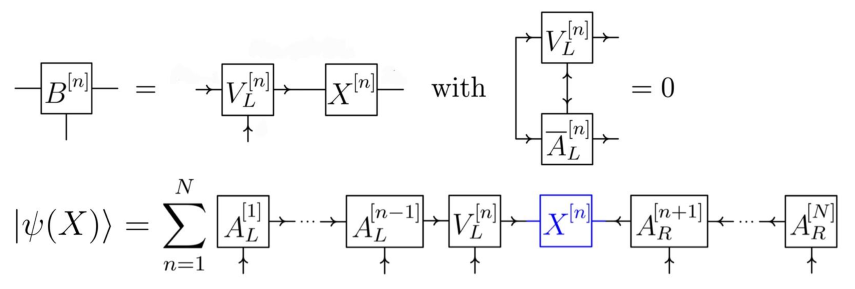
Ground state

Tangent space



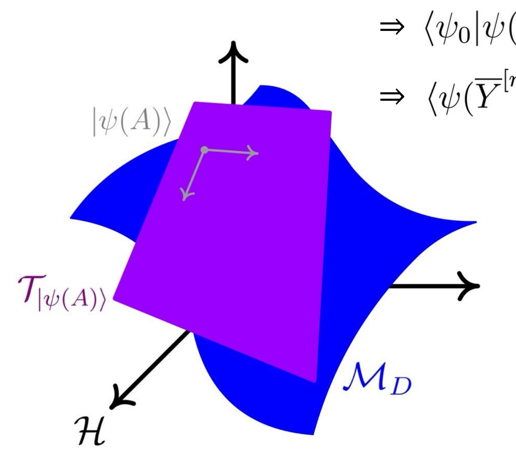
Gauge freedom $B^{[n]} \to B^{[n]} + A_L^{[n]} Y^{[n]} - Y^{[n-1]} A_L^{[n]}$

Parametrization $B(X)$ such that (i) $\langle\psi_0|\psi(X)\rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective

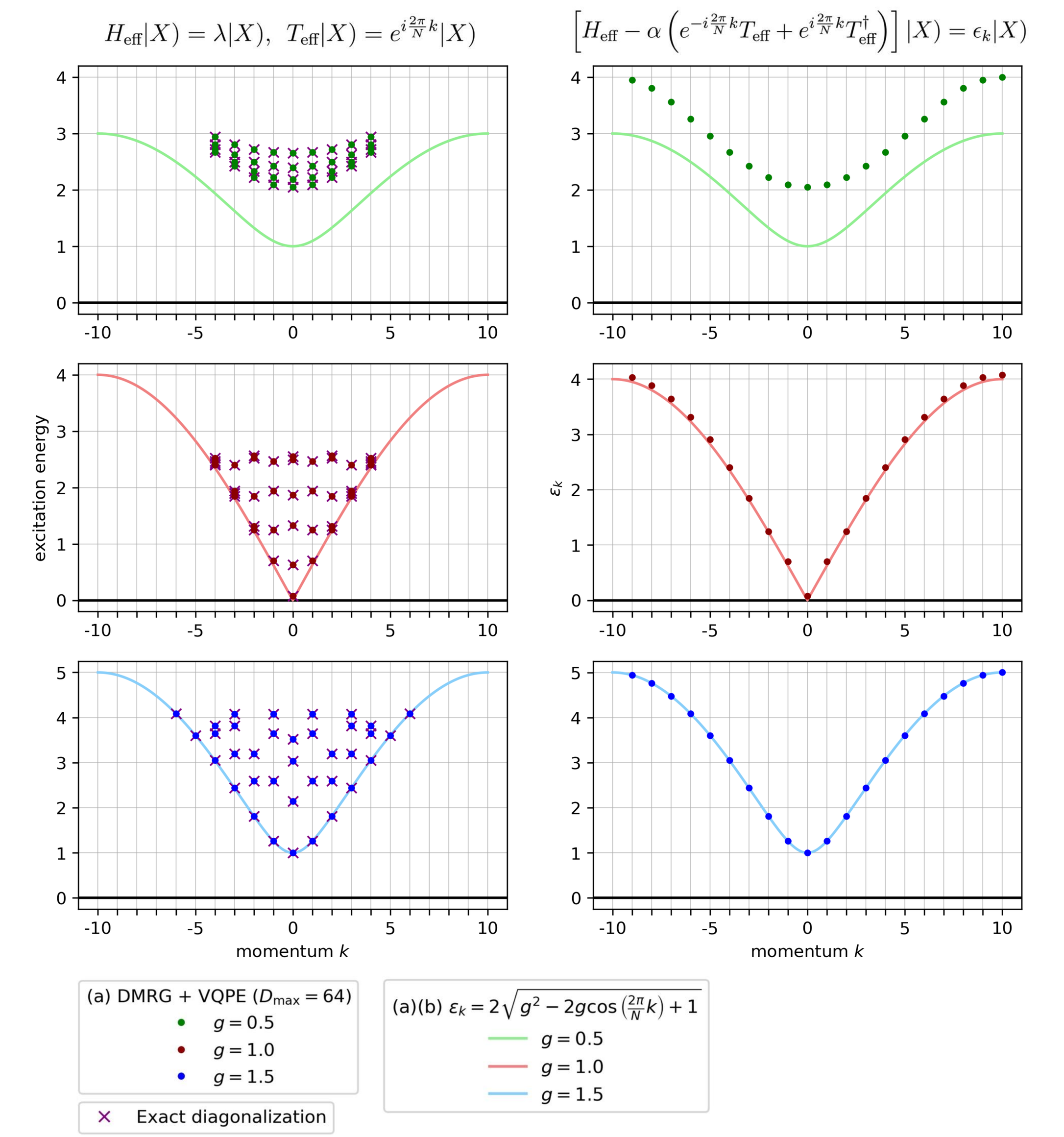


$\Rightarrow \langle\psi_0|\psi(X^{[n]})\rangle = 0$

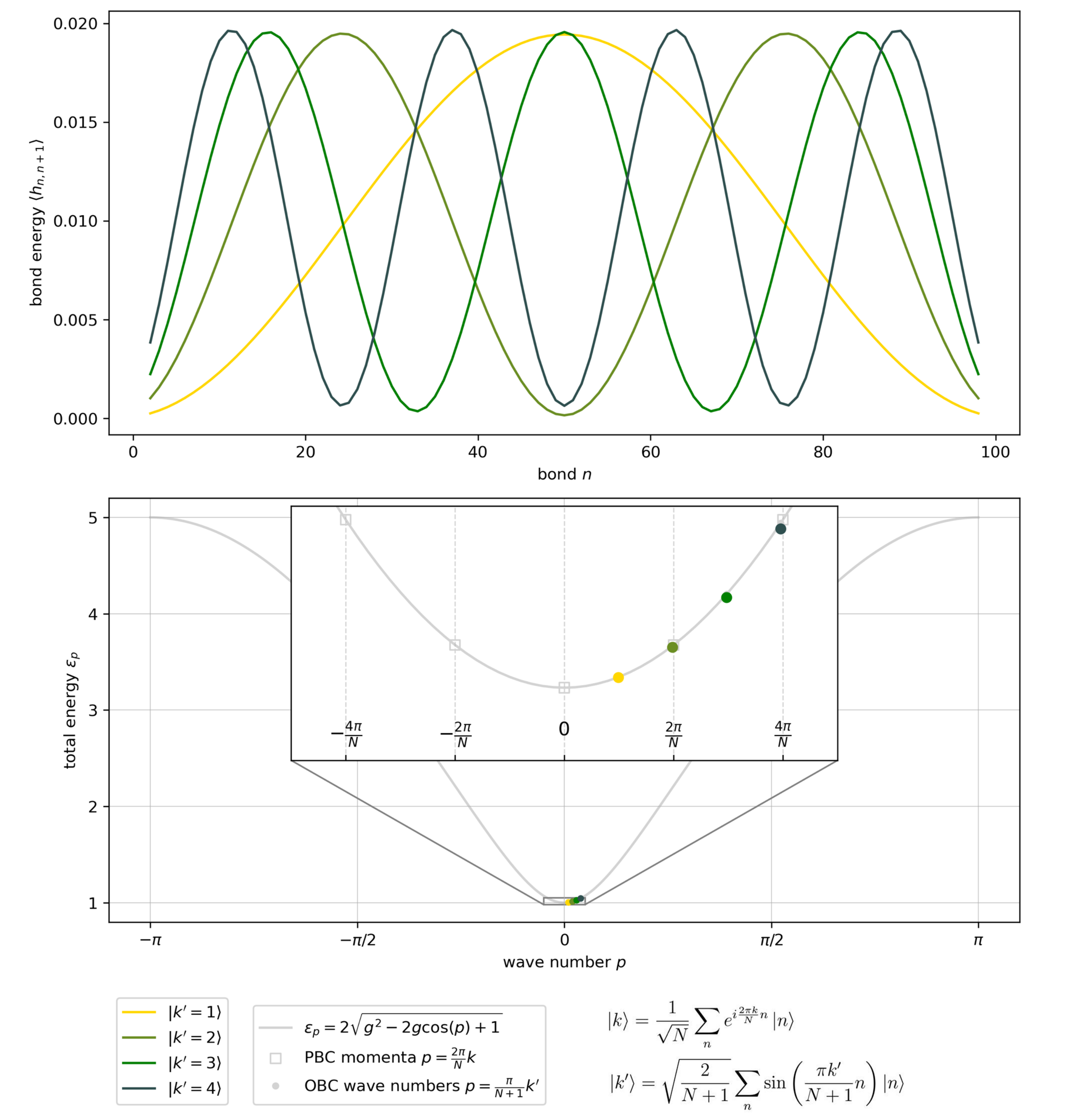
$\Rightarrow \langle\psi(\overline{Y}^{[m]})|\psi(X^{[n]})\rangle = \delta_{m,n}(\overline{Y}^{[n]}|X^{[n]})$



Restoring momentum with PBC Hamiltonian

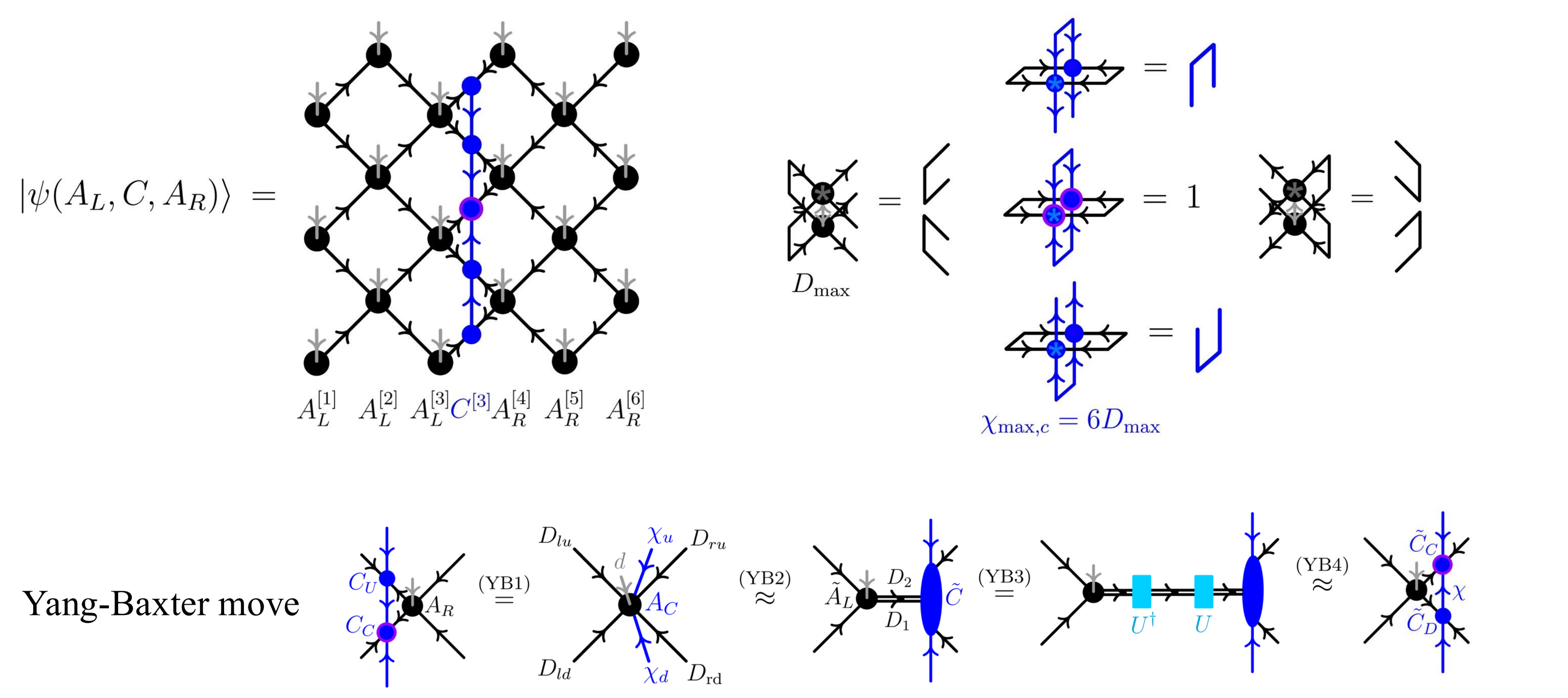


Standing waves with OBC Hamiltonian

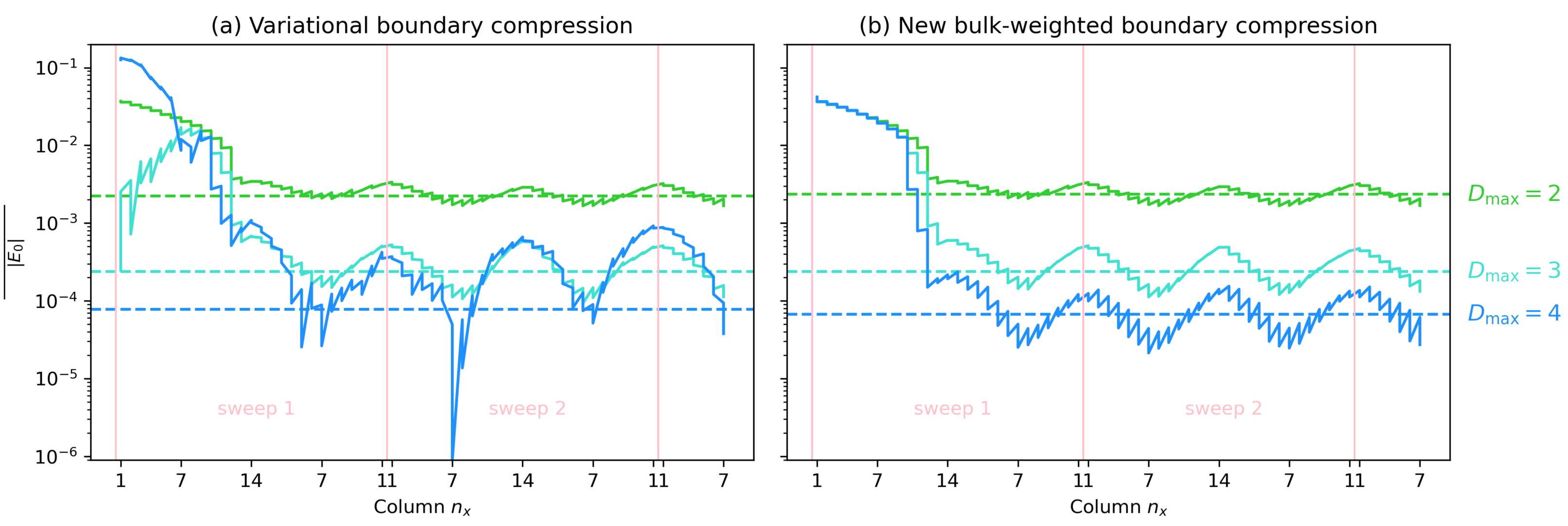


isoTNS

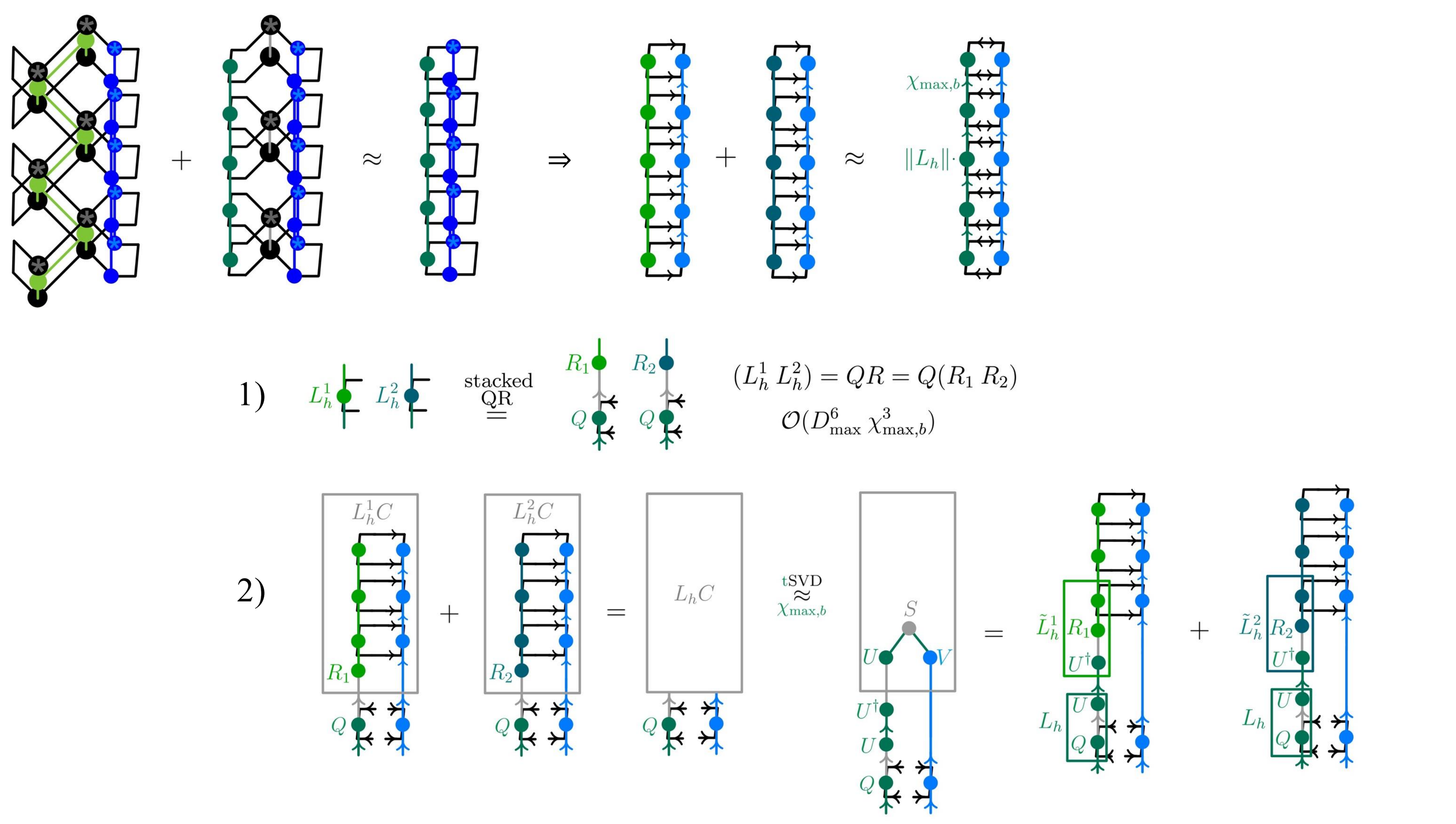
Ground state ansatz



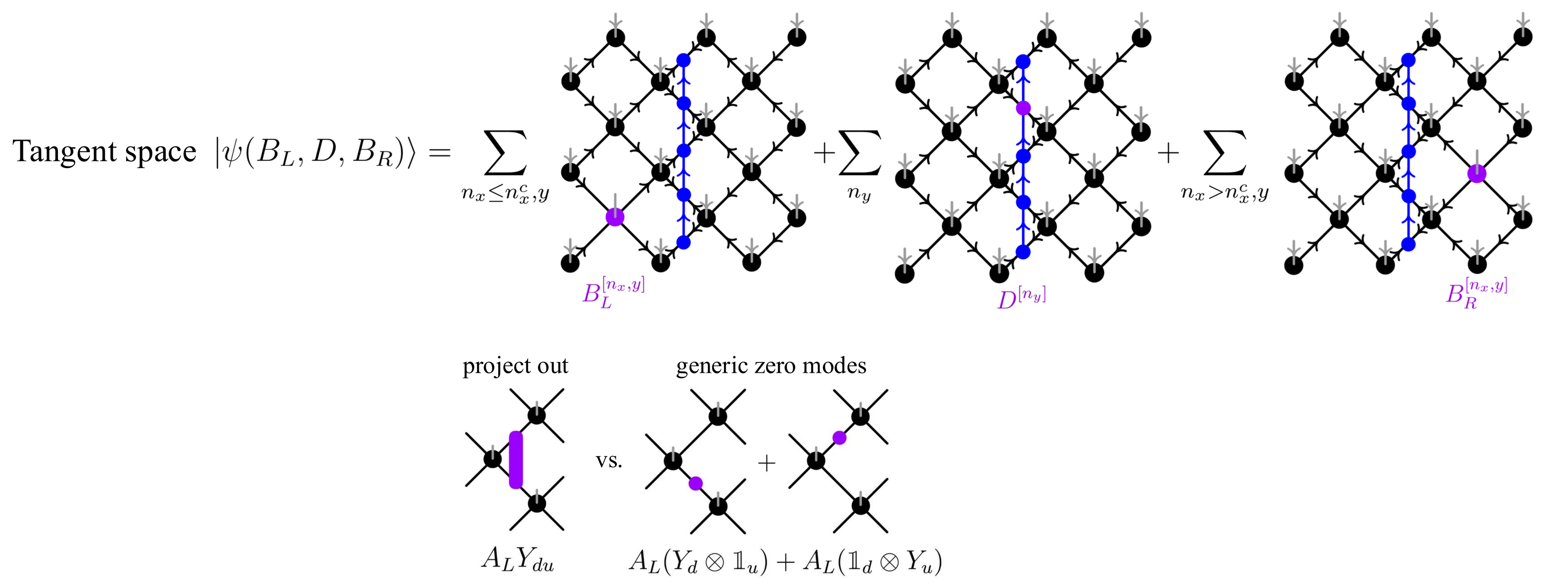
DMRG²



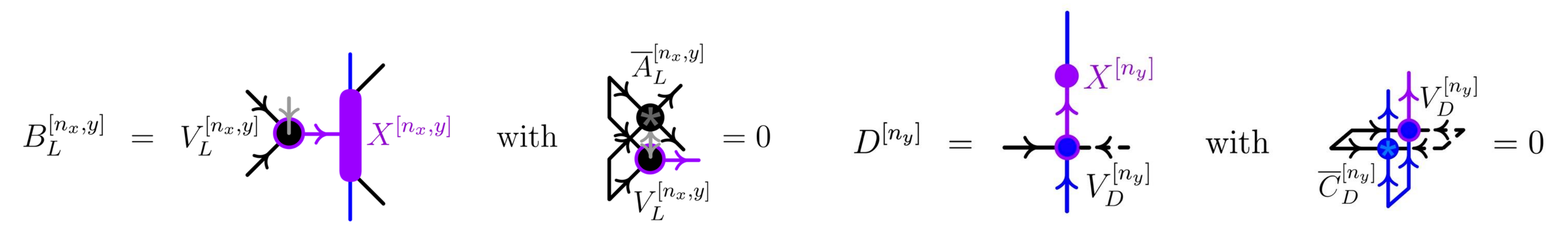
New bulk-weighted boundary compression



New quasiparticle excitation ansatz



Parametrization $B_L(X), D(X), B_R(X)$ such that (i) $\langle\psi_0|\psi(X)\rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective



$\Rightarrow \langle\psi_0|\psi(X^{[n_x,y]})\rangle = 0$

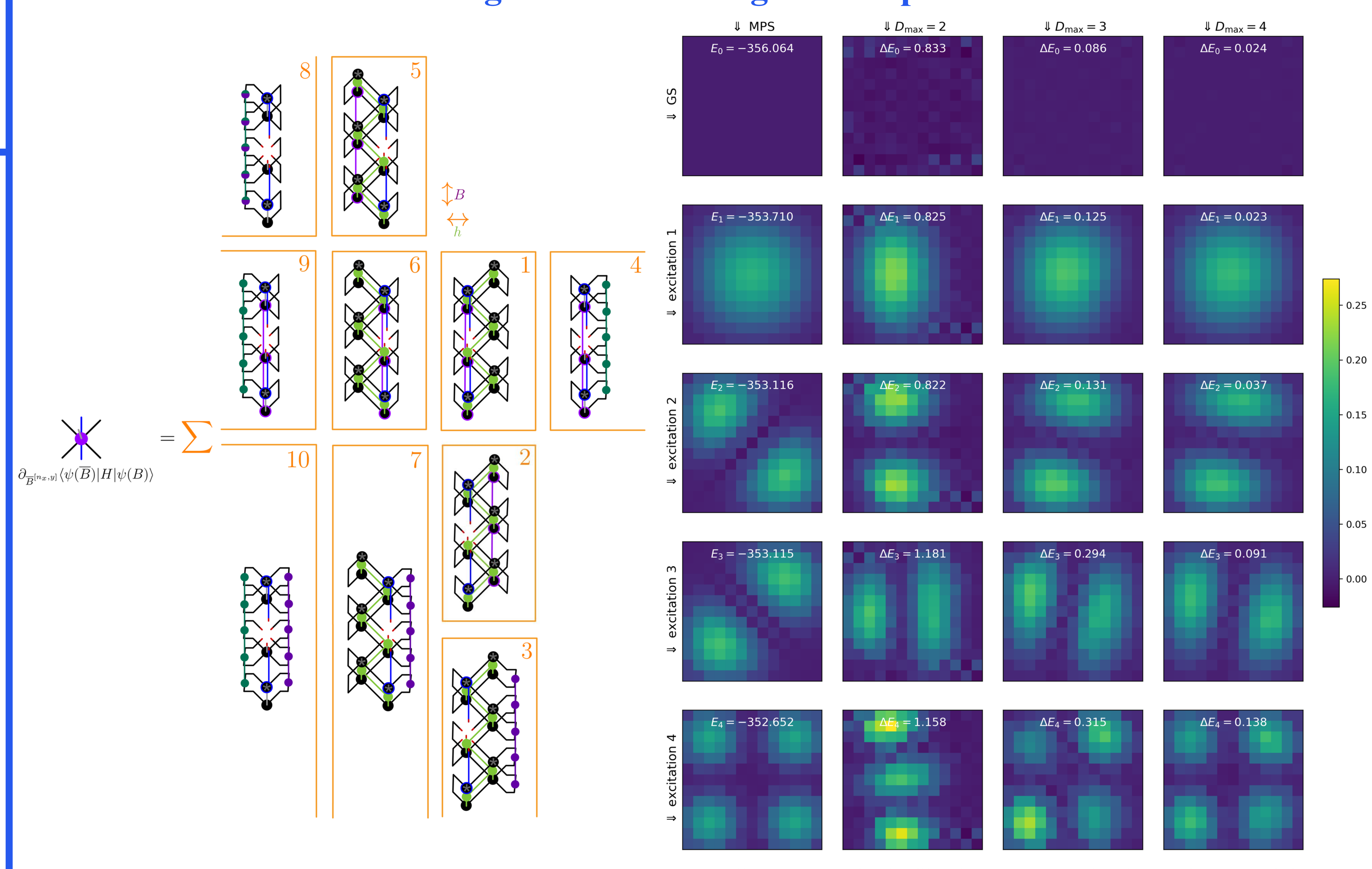
$\Rightarrow \langle\psi(\overline{X}^{[n_y]})|\psi(X^{[n_x,y]})\rangle = 0$

$\Rightarrow \langle\psi(\overline{X}^{[\tilde{n}_x,\tilde{y}]})|\psi(X^{[n_x,y]})\rangle = \delta_{\tilde{n}_x,n_x}\delta_{\tilde{y},y}(\overline{X}^{[n_x,y]}|X^{[n_x,y]})$

$\Rightarrow \langle\psi_0|\psi(X^{[n_y]})\rangle = 0$

$\Rightarrow \langle\psi(\overline{X}^{[\tilde{n}_y]})|\psi(X^{[n_y]})\rangle = \delta_{\tilde{n}_y,n_y}(\overline{X}^{[n_y]}|X^{[n_y]})$

Solving the standard eigenvalue problem



Standing waves of spin flips in paramagnetic phase of TFI model

References

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[3] Sappler et al., Diagonal Isometric Form for Tensor Product States in Two Dimensions, 2025

[4] Haegeman et al., Post-matrix product state methods: To tangent space and beyond, 2013

[5] Van Damme et al., Efficient matrix product state methods for extracting spectral information on rings and cylinders, 2021

[6] Wittmann, https://github.com/lukasjwittmann/iso_tns-public, 2025